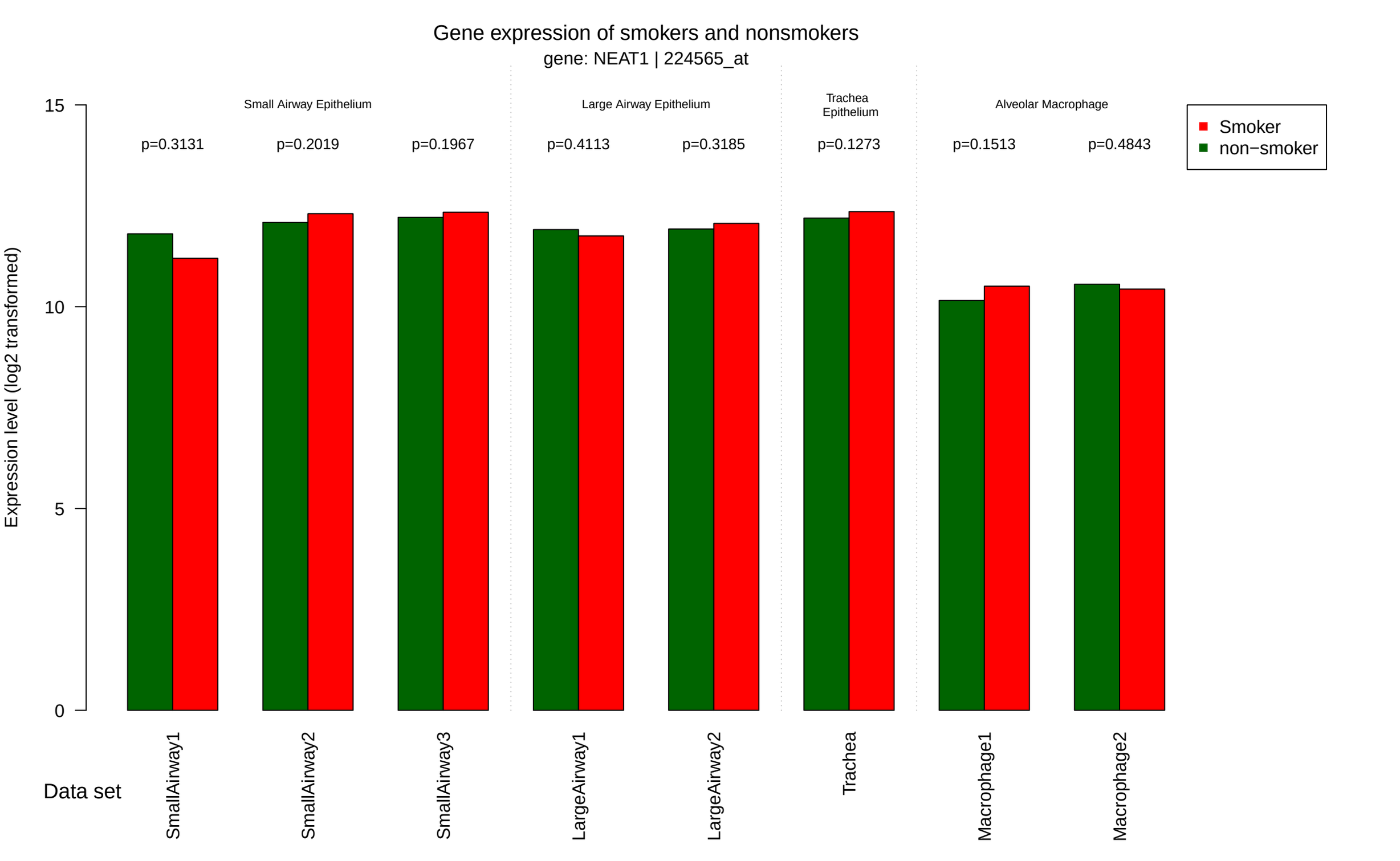
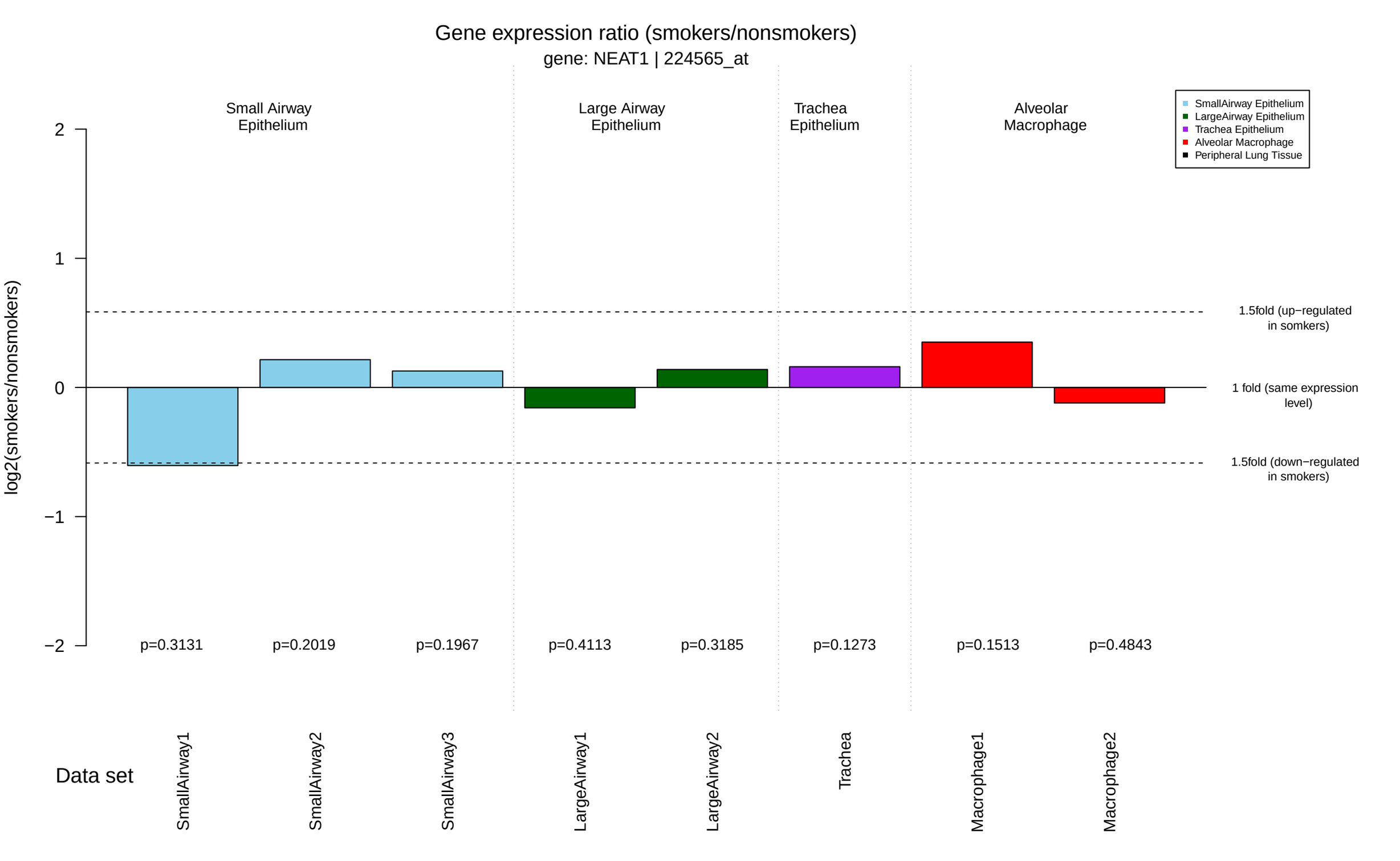
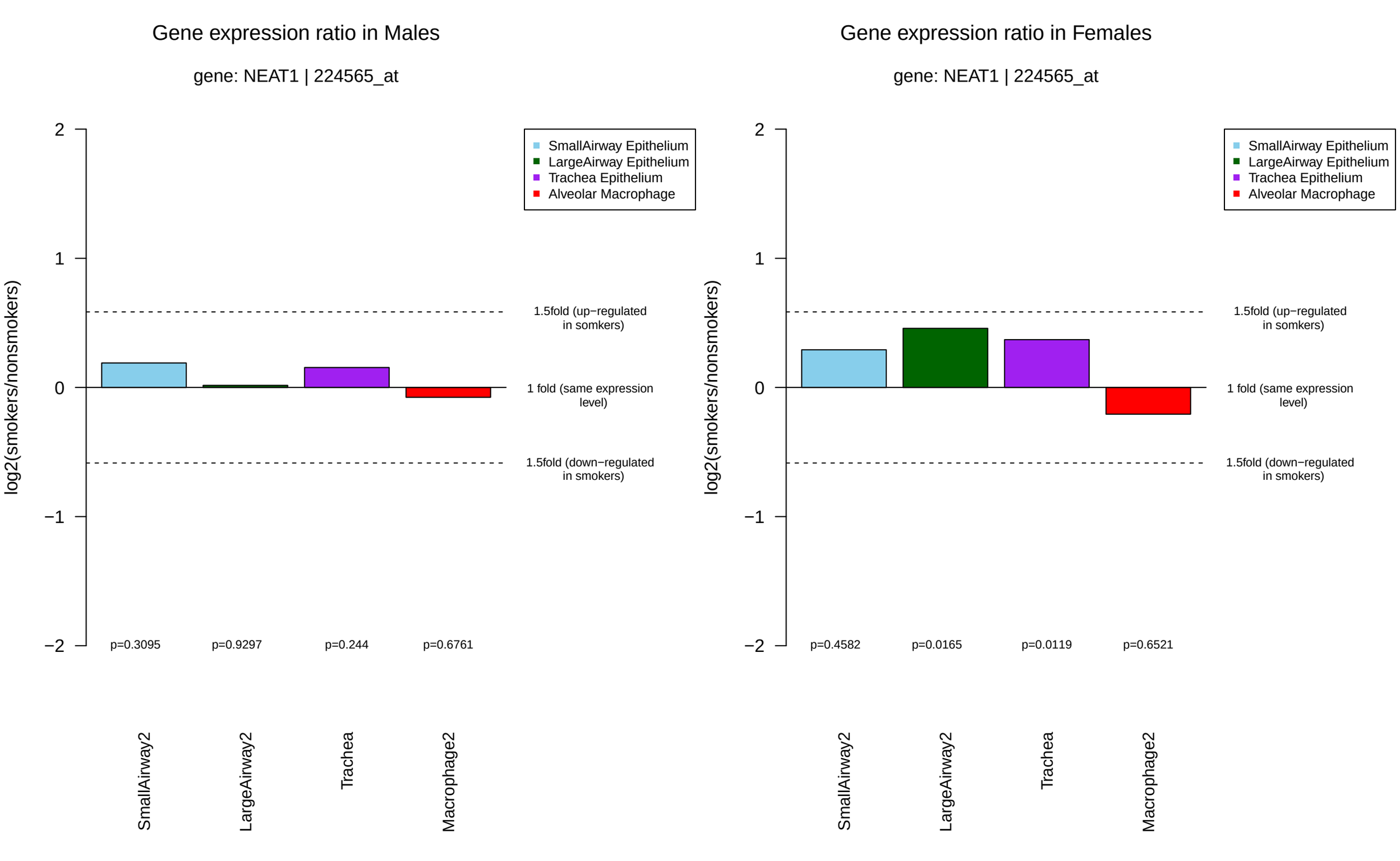
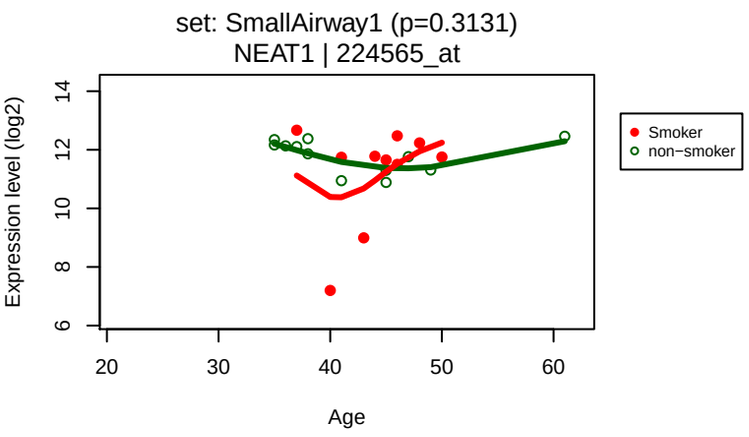
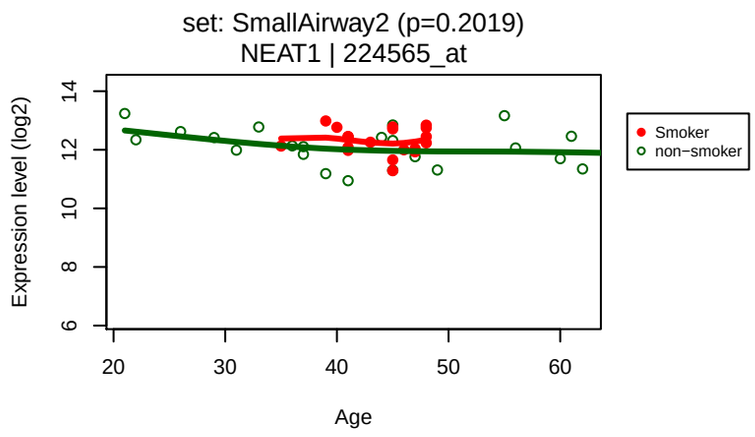
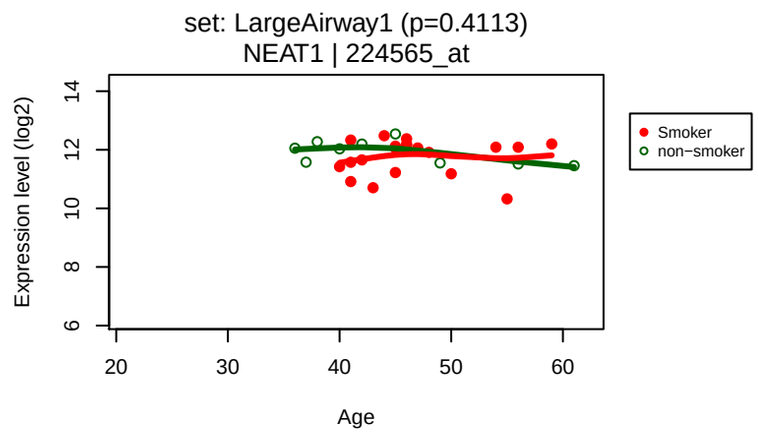

Gene expression of smokers and nonsmokers
gene: NEAT1 | 224565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3131
p=0.2019
p=0.1967
p=0.4113
p=0.3185
p=0.1273
p=0.1513
p=0.4843
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

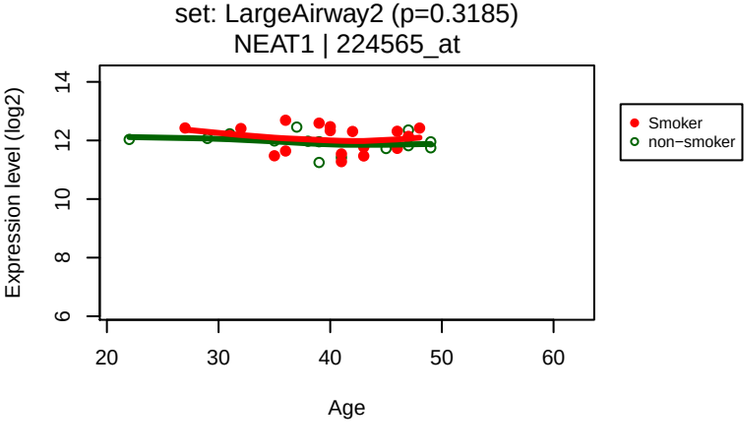

Gene expression ratio (smokers/nonsmokers)
gene: NEAT1 | 224565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3131
p=0.2019
p=0.1967
p=0.4113
p=0.3185
p=0.1273
p=0.1513
p=0.4843
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

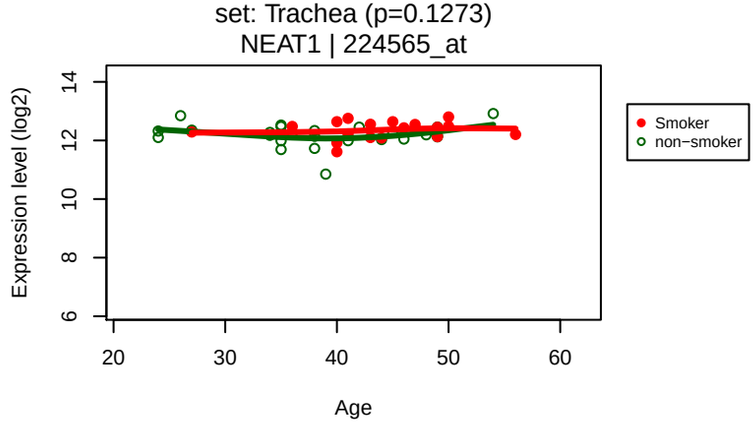

Gene expression ratio in Males
gene: NEAT1 | 224565_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3095
p=0.9297
p=0.244
p=0.6761
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEAT1 | 224565_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4582
p=0.0165
p=0.0119
p=0.6521
SmallAirway2
LargeAirway2
Trachea
Macrophage2

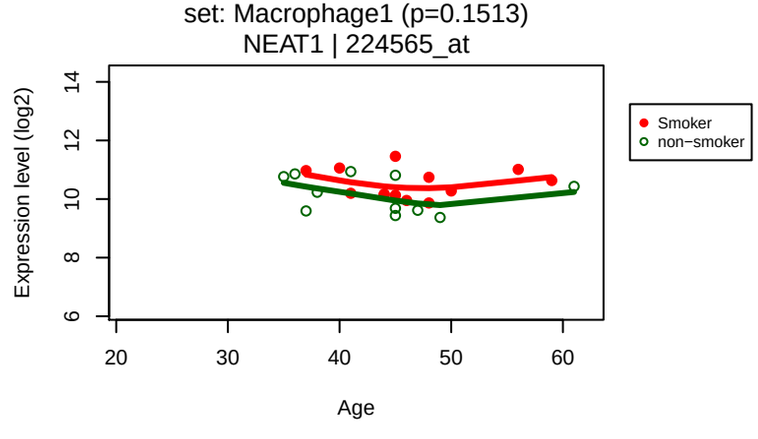

set: SmallAirway1 (p=0.3131)
NEAT1 | 224565_at
Expression level (log2)
Age
Smoker
non−smoker

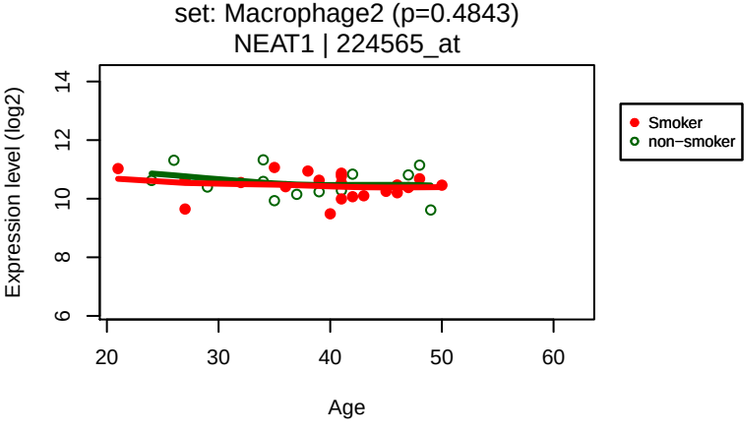

set: SmallAirway2 (p=0.2019)
NEAT1 | 224565_at
Expression level (log2)
Age
Smoker
non−smoker

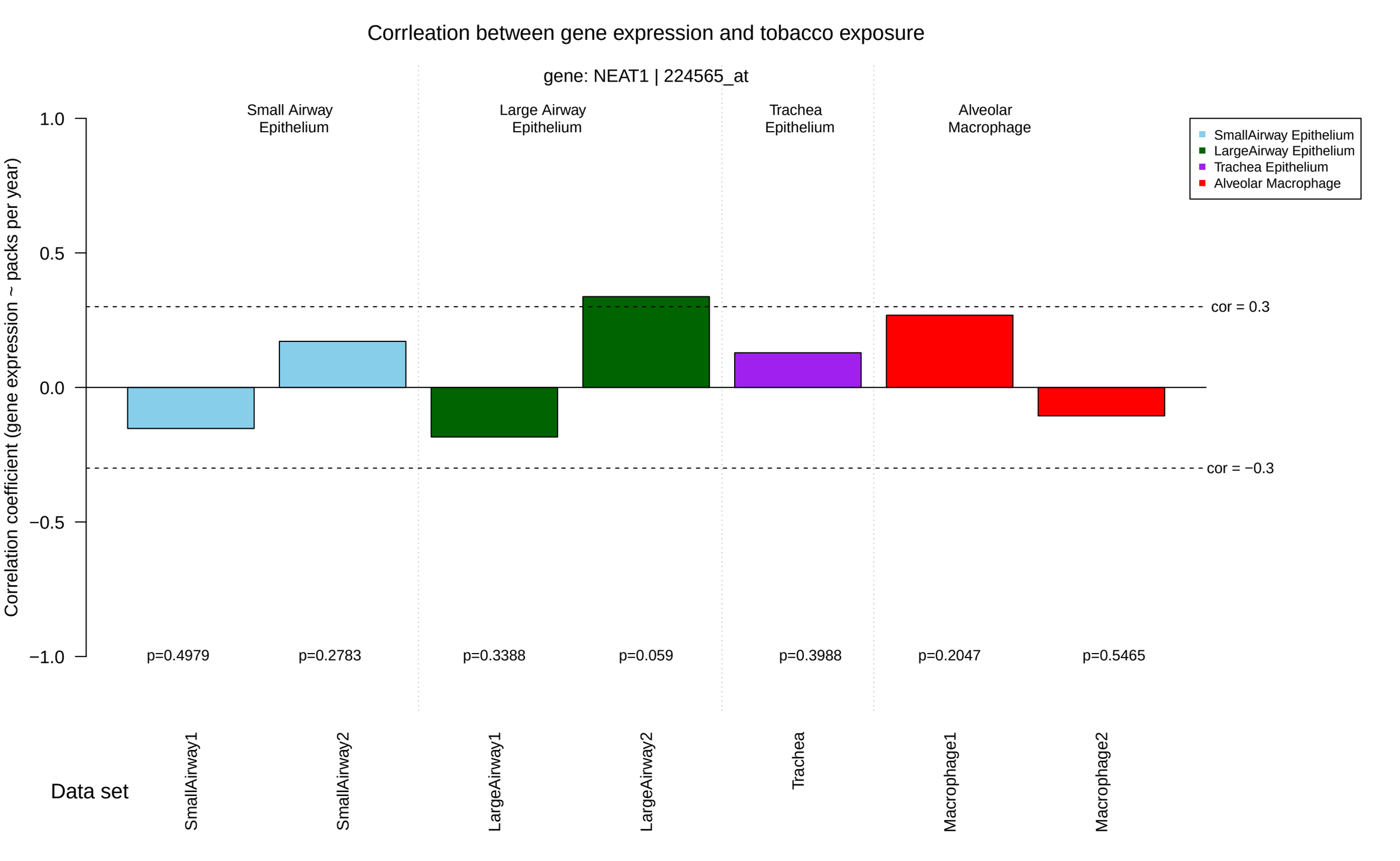

set: LargeAirway1 (p=0.4113)
NEAT1 | 224565_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3185)
NEAT1 | 224565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1273)
NEAT1 | 224565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1513)
NEAT1 | 224565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4843)
NEAT1 | 224565_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEAT1 | 224565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4979
p=0.2783
p=0.3388
p=0.059
p=0.3988
p=0.2047
p=0.5465
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2